

Correspondance projet vs cours

1. Python Dynamic Typing Problem

- ✓ Présent dans le projet
 - ☞ Montres le problème indirectement avec :
 - erreurs de types (string au lieu de int)
 - validation Pydantic qui bloque les erreurs
 - ✓ Donc c'est bien couvert
-

2. How to Use Pydantic

- ✓ Oui totalement

voici :

- BaseModel
 - création de modèles (User, Product, Order)
 - instanciation d'objets
-

3. Validating Data with Pydantic

- ✓ OUI (très bien fait dans le projet)

Teste :

- int vs string
- email invalide
- erreurs automatiques Pydantic

- ☞ exactement comme dans le cours
-

4. Custom Field Validation

- ✓ OUI :
 - field_validator
 - règle : account_id > 0
 - ☞ correspond 100% au cours
-

5. JSON Serialization

✓ OUI

Utilises :

- `model_dump()`
- `model_dump_json()`

👉 exactement ce que demande le cours

6. Pydantic vs Dataclasses

✓ OUI :

- comparaison dans output
 - explication validation vs pas validation
-

🏆 Conclusion claire

Partie du cours	mon projet
Dynamic typing problem	✓ couvert indirectement
BaseModel usage	✓ oui
Validation	✓ oui
Custom validation	✓ oui
JSON serialization	✓ oui
Dataclass comparison	✓ oui

Screen

Shoot :

```
Windows PowerShell
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Installez la dernière version de PowerShell pour de nouvelles fonctionnalités et améliorations ! https://aka.ms/PSWindows

PS D:\python_avancé_2SE\TPs\TP1_Pydantic> python -m venv venv
PS D:\python_avancé_2SE\TPs\TP1_Pydantic> venv\Scripts\activate
(venv) PS D:\python_avancé_2SE\TPs\TP1_Pydantic> pip install pydantic[email]
Collecting pydantic[email]
  Downloading pydantic-2.13.3-py3-none-any.whl.metadata (108 kB)
Collecting annotated-types>=0.6.0 (from pydantic[email])
  Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.3 (from pydantic[email])
  Downloading pydantic_core-2.46.3-cp314-cp314-win_amd64.whl.metadata (6.7 kB)
Collecting typing-extensions>=4.14.1 (from pydantic[email])
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from pydantic[email])
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting email-validator>=2.0.0 (from pydantic[email])
  Using cached email_validator-2.3.0-py3-none-any.whl.metadata (26 kB)
Collecting dnspython>=2.0.0 (from email-validator>=2.0.0->pydantic[email])
  Using cached dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Collecting idna>=2.0.0 (from email-validator>=2.0.0->pydantic[email])
  Downloading idna-3.13-py3-none-any.whl.metadata (8.0 kB)
Downloading pydantic-2.13.3-py3-none-any.whl (471 kB)
Downloading pydantic_core-2.46.3-cp314-cp314-win_amd64.whl (2.1 MB)
  2.1/2.1 MB 20.5 kB/s 0:01:36
Using cached annotated_types-0.7.0-py3-none-any.whl (13 kB)
```

```
Windows PowerShell X + v - □ X

[notice] A new release of pip is available: 25.3 -> 26.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS D:\python_avancé_2SE\TPs\TP1_Pydantic> pip freeze > requirements.txt
(venv) PS D:\python_avancé_2SE\TPs\TP1_Pydantic> pip install -r requirements.txt
Requirement already satisfied: pydantic>=2.0.0 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from -r requirements.txt (line 1)) (2.13.3)
Requirement already satisfied: email-validator>=2.0.0 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from -r requirements.txt (line 2)) (2.3.0)
Collecting pytest>=7.0.0 (from -r requirements.txt (line 3))
  Using cached pytest-9.0.3-py3-none-any.whl.metadata (7.6 kB)
Requirement already satisfied: annotated-types>=0.6.0 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from pydantic>=2.0.0->-r requirements.txt (line 1)) (0.7.0)
Requirement already satisfied: pydantic-core==2.46.3 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from pydantic>=2.0.0->-r requirements.txt (line 1)) (2.46.3)
Requirement already satisfied: typing-extensions>=4.14.1 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from pydantic>=2.0.0->-r requirements.txt (line 1)) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.2 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from pydantic>=2.0.0->-r requirements.txt (line 1)) (0.4.2)
Requirement already satisfied: dnspython>=2.0.0 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from email-validator>=2.0.0->-r requirements.txt (line 2)) (2.8.0)
Requirement already satisfied: idna>=2.0.0 in d:\python_avancé_2se\tps\tp1_pydantic\venv\lib\site-packages (from email-validator>=2.0.0->-r requirements.txt (line 2)) (3.13)
Collecting colorama>=0.4 (from pytest>=7.0.0->-r requirements.txt (line
```

1. DEMONSTRATION DE BASEMODEL :

```
Windows PowerShell
(venv) PS D:\python_avancé_2SE\TPs\TP1_Pydantic> python -m src.main

=====
TP1 PYDANTIC - DEMONSTRATION
=====

1. DEMONSTRATION DE BASEMODEL
=====

User cree avec succes :
- name: Salah
- email: salah@example.com
- account_id: 12345
- age: 25
=====
```

2. DEMONSTRATION DE LA VALIDATION :

```
Windows PowerShell

=====
2. DEMONSTRATION DE LA VALIDATION
=====

Erreur (account_id invalide) : 1 validation error for User
account_id
  Input should be a valid integer, unable to parse string as an integer
[type=int_parsing, input_value='hello', input_type=str]
  For further information visit https://errors.pydantic.dev/2.13/v/int_parsing

Erreur (email invalide) : 1 validation error for User
email
  value is not a valid email address: An email address must have an @-sign.
[type=value_error, input_value='ali', input_type=str]

Erreur (account_id negatif) : 1 validation error for User
account_id
  Value error, account_id must be positive: -12 [type=value_error, input_value=-12, input_type=int]
  For further information visit https://errors.pydantic.dev/2.13/v/value_error
=====
```

3. SERIALISATION JSON :

```
Windows PowerShell

=====
3. SERIALISATION JSON
=====
Original: name='Ali' email='ali@example.com' account_id=1234 age=30
JSON: {"name":"Ali","email":"ali@example.com","account_id":1234,"age":30}
Dict: {'name': 'Ali', 'email': 'ali@example.com', 'account_id': 1234, 'age': 30}
```

4. MODELE COMPLEXE (ORDER) :

```
Windows PowerShell

=====
4. MODELE COMPLEXE (ORDER)
=====
Commande #5001 - Total: 1059.97
```

5. PYDANTIC VS DATACLASSES :

```
Windows PowerShell

=====
5. PYDANTIC VS DATACLASSES
=====
Pydantic: validation automatique
Dataclass: pas de validation automatique

=====
FIN
=====
```

